

Deliverable 5 - Streamlit Mock-up

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1. Purpose of the mock-up

The purpose of this mock-up is to show the look and feel of the future Streamlit application. It is not yet a final application connected to the complete model.

2. What the prototype shows

The prototype contains the following sections:

- project overview;
- simulated input form for wine characteristics;
- mocked prediction;
- presentation of the three models;
- display of the performance chart if available;
- next steps to move from mock-up to real application.

3. Mock inputs

The form contains fields for the physicochemical variables: acidity, residual sugar, chlorides, sulfur dioxide, density, pH, sulphates, alcohol, and wine type. These fields show how a user could enter information about a wine.

4. Mock prediction

The prediction is intentionally simulated. The button displays for example Predicted quality group: medium quality. The application clearly states that the prediction is mocked and that the final model is not connected yet.

5. Plan to replace mock inputs with final outputs

Plan to replace the mock-up with a real version:

- load the final preprocessing;
- load the selected model;
- transform the user inputs;
- generate a real prediction;
- display the predicted class;
- optionally display probabilities;
- integrate the final plots and metrics.

6. Screenshots

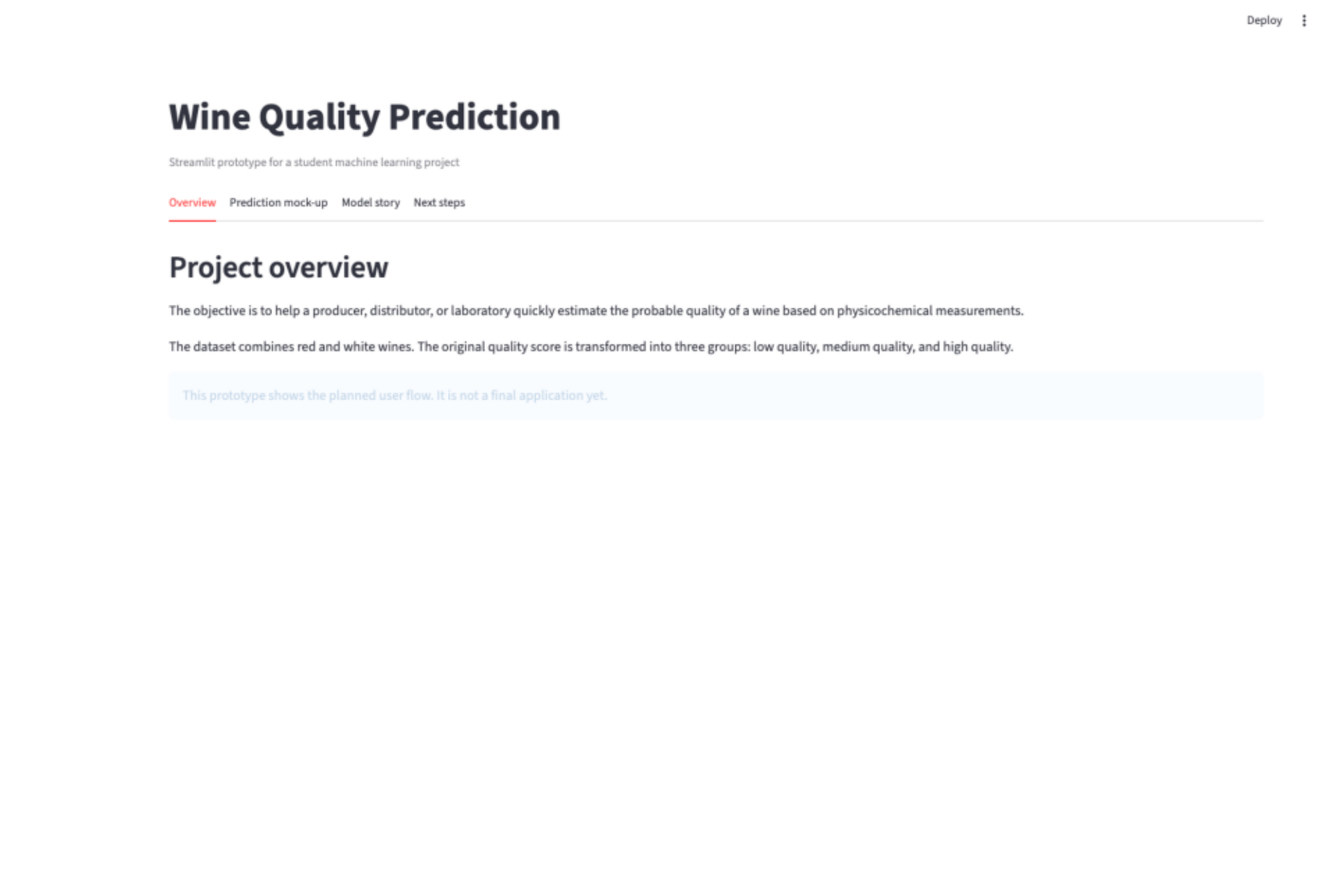
The screenshots were generated automatically with Playwright after launching Streamlit:

- screenshots/streamlit_mockup_home.png;
- screenshots/streamlit_mockup_prediction.png;
- screenshots/streamlit_mockup_model_story.png.

Command used to launch the application:

```
python3 -m streamlit run src/app.py --server.headless true --server.port 8501  
--server.address 127.0.0.1
```

Screenshot 1 - Streamlit home



Home / project overview screen.

Screenshot 2 - Mock prediction

Deploy

Wine Quality Prediction

Streamlit prototype for a student machine learning project

Overview

Prediction mock-up

Model story

Next steps

Mock input form

fixed acidity 7.00 - +	total sulfur dioxide 115.00 - +
volatile acidity 0.30 - +	density 0.995 - +
citric acid 0.30 - +	pH 3.20 - +
residual sugar 5.00 - +	sulphates 0.50 - +
chlorides 0.050 - +	alcohol 10.50 - +
free sulfur dioxide 30.00 - +	wine_type white ▼

Predict

Mock input and mocked prediction screen.

Wine Quality Prediction

Streamlit prototype for a student machine learning project

Overview Prediction mock-up **Model story** Next steps

Model story

Logistic Regression

A simple model used as a clear and easy-to-explain baseline.

Random Forest

A more flexible model, useful for non-linear relationships in tabular data.

SVM

A model based on class separation, interesting with standardized variables.

Model performance

